Presentation 2

• Links to presentation(s) and code(s) on GitHub:

- Presentation 2
- Dashboard
- Code

• What did you do?

I first re-ran the CAR import data code to include the new call data from the month of September. Using that dataset in python, I ensured that the activity timestamps were in a proper datetime format to allow accurate time-based analysis. Then, I grouped the data by each contact session to calculate key call metrics: the earliest timestamp of the session (Call_Start_Time), the starting hour of the call, the number of unique activity timestamps within the session, and the total call duration in minutes. Finally, I created separate columns for the year, month, and day of the week for each call, enabling me to create a hierarchical filter in PowerBI.

I then grouped the call data by each contact session to summarize key indicators, including whether a session had a CallbackRetry, a LegalServerScreenPop, or an assigned agent. I then applied a logical rule to infer whether the call was likely picked up: sessions with a CallbackRetry and no agent were marked as not picked up, while sessions with either a LegalServerScreenPop or an agent were marked as picked up. Afterward, I checked for contradictions where the observed activity patterns did not align with this assumption, such as sessions containing both CallbackRetry and LegalServerScreenPop or CallbackRetry alongside an agent, and compiled those cases for inspection, ultimately printing the total sessions, the number of contradictions found, and a preview of those contradictory sessions.

In PowerBI, I put the Starting Hour on the x-axis filtering for times from 8-17, and Count of EP Name, filtering by Courtesy Callback Telephony EP. I added the Picked Up column as the legend. I then created a slicer to include year, month, then day of the week.

• How does it help the project?

My analysis helps Legal Aid by identifying patterns in callback timing and pickup rates, revealing that the current scheduling process is misaligned with customer availability. The analysis reveals that a large share of callbacks occur early in the morning when pickup rates are low, and that unanswered callbacks exceed completed ones across all hours. These insights provide a data-driven basis for improving the callback process through better scheduling, smarter retry logic, and potentially implementing confirmation or rescheduling options.

• Issues faced, Attempts to resolve issues, and Issues resolved

Determining if a callback was picked up → resolved by determining that CallbackRetry and LegalServerScreenPop in combination with presence of agent name could serve as an indication of callback pickup. Need to clarify this with Cynthia however.

• Next steps

My next steps will be to analyze agent-level performance with the objective of determining if certain agents have higher callback pickup rates and uncovering patterns in missed callbacks. To achieve this, I will visualize pickup performance by agent by building a stacked bar chart in Power BI that displays the number of callbacks picked up versus those missed for each agent. This approach will allow me to identify performance trends, highlight agents who consistently achieve higher pickup rates, and pinpoint areas where additional support or training may be beneficial.

• References (Mention if you built up on someone else's work)

Used the CAR_data_import_code file to combine the datasets, created by Prof. Krishna.

Included the new September 2025 data.

Presentation 1

• Links to presentation(s) and code(s) on GitHub:

- Google Slides Presentation 1
  - See Julia Schaffner's documentation for a link to the ppt presentation with PowerBI embedded visualizations. The google slides only contain the static graphics.
- Dashboard 1
  - All coding was done in Power BI

• What did you do?

I collaborated with Julia in order to create the visualization "No Answer From Customer" count versus "Starting Hour" bar graph. I first started by utilizing the code provided by Prof. Krishna to compile the All Calls Data by month into one single csv file. I also ran the code once again provided by Prof. Krishna to compile the CAR flow into a singular csv file. I then uploaded the data to PowerBI, but ultimately chose to work within Julia's PowerBI dashboard for efficiency and timing reasons.

CAR Data:

To prepare the data for further analysis in Power BI, Julia first used the "Group By" function to ensure that each *Contact Session ID* corresponded to a single row, preventing double counting. She then applied the "Advanced Group By" function to perform multiple aggregations, creating four new columns: *Count*, *Call Start Time*, *Starting Hour*, and *All Rows*. Finally, Julia used the "Expand All Rows" feature to access and visualize the nested information within the dataset.

I then plotted the *Starting Hour* against the count of *EP name* filtering to only the *Courtesy Callback Telephony EP* to visualize call activity patterns of callbacks throughout the day. I filtered the data to include only working hours, from 8:30 a.m. to 5:00 p.m., using solely the CAR data source for this analysis. To refine the results, I applied the *Termination Reason* field as a legend, selecting only cases labeled "agent left" and "customer left" This allowed me to focus specifically on unsuccessful callbacks (where either the agent or customer left the call) within regular business hours.

I additionally plotted *Starting Hour* against the count of *Agent Name* to visualize the number of agents available throughout the day. I once again filtered the data to include only working hours, from 8:00 a.m. to 5:00 p.m. I did not include the rows that *Agent Name* was blank.

• How does it help the project?

This analysis helps the project by revealing that customers are leaving at the same rate agents are leaving despite having a relatively consistent amount of agents each hour of the day. When the callback volume is the highest in the first hour, we see the majority of customers leaving the call which makes sense. However when the call volume decreased, customers were leaving the calls at a proportional rate to the first hour of the day. I compared this to the number of agents staffed at each hour of the day. While most drop-offs occur during the first hour of high callback volume, the continued leave rates later cannot be explained by the slight decrease in agents. This suggests that factors like agent responsiveness, rather than staffing levels alone, are driving customer drop-offs and should be the focus of improvements.

• Issues faced, Attempts to resolve issues, and Issues resolved

The total number of callbacks was 72,426, of which 44 were terminated because the customer did not answer. Since termination reasons are only recorded for calls made after March 16, 2025, the analysis focuses on this subset of 11,916 calls. This timeframe is a drawback, only allowing me to explore 17% of the data, however it is also beneficial because it provides clearer insight into why certain callbacks are not being completed. However, with only 44 instances of customers not answering, this group is too small to generate actionable recommendations hence why I switched from my original recommendation. Therefore, the analysis was expanded to include termination reasons that represent larger proportions of the overall dataset such as "agent left" and "customer left".

• Next steps

- Segment by call type or customer profile
    - Identify whether certain types of calls or customer groups are more likely to leave, which could reveal certain patterns
- Refine dashboard using other filters to find if other EP names follow a similar hourly pattern

• References (Mention if you built up on someone else's work)

Used the CAR_data_import_code file and AllCallsData_import_code to combine the datasets, created by Prof. Krishna.